

Book 7: Egg Quality, Grading & Post-Harvest Handling

A Comprehensive Operational Guide to Shell Integrity, Internal Diagnostics & Storage Optimization

PoultryWise Operational Manuals • Post-Harvest Quality & Processing Standard

TABLE OF CONTENTS

Chapter 1: Anatomy & Physiology of Egg Formation	Chapter 6: Washing, Sanitizing & Shell Coating Protocols
Chapter 2: Internal Egg Quality Assessment & Haugh Units	Chapter 7: Cold Chain Logistics & Storage Microbiology
Chapter 3: External Quality & Shell Strength Metrics	Chapter 8: Egg Packaging, Cartoning & Transport Stress
Chapter 4: Identifying & Resolving Shell Defects	Chapter 9: Value-Added Liquid & Powdered Egg Processing
Chapter 5: Automated Candling & Size Grading Systems	Chapter 10: Quality Assurance & Regulatory Compliance

CHAPTER 1: ANATOMY & PHYSIOLOGY OF EGG FORMATION

Understanding the 24-to-26 hour biological cycle of egg development in laying hens is critical for diagnosing shell quality issues and optimizing layer nutrition.

- **Ovary:** Contains thousands of ovocytes. Follicular maturation releases a yolk into the oviduct roughly 30 minutes after the previous oviposition.
- **Oviduct Segments:**
 - *Infundibulum (15–30 mins):* Captures the yolk; site of fertilization in breeding stock.
 - *Magnum (3 hours):* Secretes thick and thin albumen (egg white) containing key proteins like ovalbumin.
 - *Isthmus (1.5 hours):* Deposits inner and outer shell membranes.
 - *Uterus / Shell Gland (18–20 hours):* Plumping phase occurs, followed by massive calcium carbonate deposition (CaCO_3) to build the shell matrix.
 - *Vagina & Cuticle Application:* Cuticle protein coating applied seconds prior to lay, sealing shell pores against bacterial entry.

CHAPTER 2: INTERNAL EGG QUALITY ASSESSMENT & HAUGH UNITS

Freshness and interior condition are measured by thick albumen height relative to egg weight, expressed in Haugh Units (HU).

Haugh Unit Formula & Quality Grades

$$HU = 100 \times \log_{10} \left(h - 1.7 \times w^{0.37} + 7.6 \right)$$

Where h is thick albumen height in millimeters and w is egg weight in grams.

- **Grade AA (HU > 72):** High firm albumen; round, raised yolk. Ideal for retail table eggs.
- **Grade A (HU 60 – 72):** Reasonably firm albumen; yolk remains centered.
- **Grade B (HU < 60):** Thin, watery albumen; flattened yolk. Best suited for commercial breaker plants.

CHAPTER 3: EXTERNAL QUALITY & SHELL STRENGTH METRICS

Shell breaking strength prevents breakage loss, which can cost farms up to 8% of total production revenue if unmonitored.

Quality Metric	Target Range	Primary Factors Influencing Metric
Shell Thickness	0.33 mm – 0.38 mm	Calcium/Phosphorus ratio, Vitamin D3 levels, hen age
Breaking Strength	> 35 N (3.5 kgf)	Shell matrix density, flock age, heat stress management
Pore Density	7,000 – 17,000 pores/egg	Uterine health, environmental temperature control
Specific Gravity	1.080 – 1.090 g/cm³	Saline flotation test standard for shell proportion

CHAPTER 4: IDENTIFYING & RESOLVING SHELL DEFECTS

Recognizing visual shell anomalies allows nutritionists and farm managers to take rapid corrective actions.

- **Soft-Shelled / Shell-less Eggs:** Caused by severe Calcium/Vitamin D3 deficiency, Infectious Bronchitis (IB), or extreme heat stress.
- **Corrugated / Misshapen Shells:** Linked to IB virus infection, stress during uterine shell deposition, or double ovulation.
- **Cracks & Micro-Fissures:** Caused by high flock age, poor cage floor design, aggressive mechanical collection, or inadequate egg washer settings.
- **Pimple-like Calcifications:** Excess calcium deposits due to shell gland disturbance or age-related mineral metabolism shifts.

CHAPTER 5: AUTOMATED CANDLING & SIZE GRADING SYSTEMS

Modern egg packing stations use high-speed automated candling (acoustic and optical sensors) to grade and sort eggs at up to 120,000 eggs/hour.

Weight Class	Minimum Weight per Egg	Minimum Weight per Dozen
Jumbo	70.9 g	850 g
Extra Large (XL)	63.8 g	765 g
Large	56.7 g	680 g
Medium	49.6 g	595 g
Small	42.5 g	510 g

CHAPTER 6: WASHING, SANITIZING & SHELL COATING PROTOCOLS

Washing eggs reduces pathogen load on the shell surface, particularly *Salmonella enteritidis*.

- **Water Temperature Rule:** Wash water must be at least 11°C warmer than the egg temperature (minimum 32°C) to prevent liquid contraction and suction of pathogens into shell pores.
- **Sanitizer Chemicals:** Chlorine (100–200 ppm) or Quaternary Ammonium Compounds applied via recirculating sprayers.
- **Mineral Oil Coating:** Food-grade mineral oil spray seals shell pores after washing, extending shelf life by reducing CO₂ loss and moisture evaporation.

CHAPTER 7: COLD CHAIN LOGISTICS & STORAGE MICROBIOLOGY

Temperature control halts microbial multiplication and slows internal degradation mechanisms.

Storage Temperature & Humidity Standards

- **Farm Holding Room:** 12°C - 15°C with 75% - 80% Relative Humidity (RH) for up to 7 days.
- **Retail Logistics & Supermarkets:** Maintain temperature continuous at ≤ 7°C (45°C ext{F}). Avoid temperature fluctuations to prevent condensation ("sweating") on eggshells.
- **Microbial Risk:** High humidity without airflow encourages mold growth (*Aspergillus*), while condensation enables *Salmonella* ingress.

CHAPTER 8: EGG PACKAGING, CARTONING & TRANSPORT STRESS

Protective packaging minimizes vibration damage and breakage along the supply chain.

- **Molded Pulp vs. Plastic Trays:** Molded pulp absorbs humidity and cushions impacts; thermoformed plastic offers higher washability and reuse in closed-loop systems.
- **Vibration Damping:** Stack height in transport vehicles should not exceed 5 pallets high; trucks require air-ride suspensions to protect against micro-fissuring.

CHAPTER 9: VALUE-ADDED LIQUID & POWDERED EGG PROCESSING

Processing surplus or Grade B eggs into liquid pasteurized products ensures zero farm waste.

- **Liquid Whole Egg Pasteurization:** Minimum heat treatment of 60°C for 3.5 minutes to eliminate pathogens without coagulating egg proteins.
- **Spray Drying:** Converts liquid egg whites or whole eggs into shelf-stable powders for industrial baking and food production.

CHAPTER 10: QUALITY ASSURANCE & REGULATORY COMPLIANCE

Implementing Hazard Analysis Critical Control Point (HACCP) guarantees food safety from flock to consumer table.

- **Critical Control Points (CCPs):**
 - CCP 1: Wash water temperature & chemical concentration monitoring.
 - CCP 2: Cold store room temperature logging ($< 7^{\circ}\text{C}$).
 - CCP 3: Swab testing for *Salmonella* on equipment contact surfaces.

End of Book 7 — PoultryWise Knowledge Series • Egg Quality, Grading & Post-Harvest Handling